RETROSPECTIVE AI RISK ASSESSMENT & GOVERNANCE PLAN

Corn FAW Detector

Agricultural AI / Computer Vision | AI Governance & Risk Management

Prepared by	Rebecca Ssesanga
Assessment type	Retrospective AI Risk Assessment
Primary framework	NIST AI Risk Management Framework (AI RMF)
Portfolio focus	Responsible AI, risk, oversight, data/model governance

Portfolio Case Study

Executive Summary

The Corn FAW Detector is a mobile machine-learning application developed during MSc research to identify visible Fall Armyworm (FAW) pest damage in maize-leaf images. The original research used 2,868 images and compared VGG16, InceptionV3 and MobileNetV2. MobileNetV2 achieved the strongest reported test performance and was converted to TensorFlow Lite for Android deployment.

Controlled testing showed strong predictive performance, but field evaluation revealed an important difference between test performance and operational risk. Of 200 field leaves predicted as Healthy, 180 were genuinely healthy while 20 were FAW affected. The application also returned Unknown when leaf structure could not be adequately detected.

Core governance conclusion

The Corn FAW Detector should be treated as an AI-enabled decision-support system, not as an autonomous agricultural diagnostic authority. Sustained deployment would require human oversight, purpose limitations, field-performance monitoring, incident management, traceability, change control and explicit accountability.

This retrospective assessment identifies 11 risks, six rated High before additional controls. It applies the NIST AI RMF functions - GOVERN, MAP, MEASURE and MANAGE - and clearly distinguishes original MSc evidence from retrospective governance findings and proposed controls.

1. Assessment Context & System Profile

The original MSc project focused on developing and evaluating a mobile ML application for FAW damage detection. This portfolio assessment asks a different question: what governance controls would be necessary if the research prototype were transitioned into sustained real-world use?

Evidence convention

Implemented during MSc research = documented technical/evaluation activity. Retrospective governance finding = conclusion from this assessment. Proposed governance control = additional measure recommended for operational deployment.

Attribute	Description
System	Corn FAW Detector
Domain	Agricultural AI / computer vision
Task	Classify maize-leaf images for visible FAW damage
ML approach	CNN transfer learning
Candidate models	VGG16, InceptionV3, MobileNetV2
Selected model	MobileNetV2
Deployment	TensorFlow Lite in Android app
Dataset	2,868 images: 1,399 FAW-damaged; 1,469 Healthy
Data geography	Bugiri and Kasese, Uganda
Potential users	Farmers, extension workers, agricultural stakeholders
Governance status	Retrospective portfolio assessment

2. Technical Evidence Base

The research used an approximately 80:10:10 train/validation/test split and augmentation including rotation, shear, zoom, horizontal flipping and width/height shifts. Three transfer-learning architectures were compared.

Model	Training accuracy	Test accuracy	F1
VGG16	97.90%	98.26%	0.98
MobileNetV2	99.87%	99.65%	1.00
InceptionV3	98.21%	92.68%	0.92

Field evaluation

- 300 leaves tested: 200 predicted Healthy, 60 predicted FAW, and 40 returned Unknown.
- Of the 200 predicted Healthy, 180 were genuinely healthy and 20 were FAW affected.
- All 60 FAW predictions matched the field assessment.
- Repeatability testing: 80 of 100 leaves produced consistent predictions.
- All 50 non-maize test images returned Unknown.

Governance implication
Strong held-out accuracy is not sufficient evidence of safe operational performance. Field false negatives, uncertainty handling, input quality and human reliance must also be governed.

3. Intended Use, Limitations & Stakeholders

Proposed intended-use statement
The Corn FAW Detector is intended to provide AI-assisted screening of maize-leaf images for visible indications of Fall Armyworm damage, supporting farmers and agricultural stakeholders in identifying cases that may require further attention or expert intervention.

Foreseeable misuse

- Treating a prediction as a definitive diagnosis or assuming Healthy guarantees absence of FAW.
- Making pest-management decisions solely from the AI output.
- Using the system for crops, pests, diseases, geographies or conditions outside its validated scope.
- Using poor-quality images without recognizing their effect on reliability.

Stakeholders

Stakeholder	Relationship	Principal concern
Farmers	Primary/end users	Incorrect or misunderstood classification
Extension workers	Expert/intermediary users	Inappropriate reliance
Agricultural experts	Verification/escalation	Quality of evidence and escalation
Researchers/developers	System improvement	Model/data limitations
Government/agencies	Potential adoption/oversight	Validation and accountability
NGOs/programmes	Potential implementation	Responsible deployment
Farming communities	Indirectly affected	Crop/economic consequences

4. Harm Analysis & Risk Method

Harm	Pathway	Potentially affected
H01 Delayed FAW response	False Healthy -> delayed investigation	Farmers
H02 Unnecessary intervention	Incorrect/over-relied result -> unnecessary action	Farmers
H03 Crop/economic impact	Detection/response failure -> avoidable crop damage	Farmers/communities
H04 Loss of trust	Repeated unreliable outcomes -> reduced confidence	Users/system owner

Risk score = Likelihood x Impact. This 5x5 method was developed for the retrospective assessment and was not part of the original MSc methodology.

Score	Classification

1-4	Low
5-9	Moderate
10-16	High
17-25	Critical

5. AI Risk Register

ID	Risk	Category	L	I	Score	Rating
R01	False-negative FAW classification	Model reliability	4	4	16	High
R02	Geographic/data generalisation	Data/generalisation	3	4	12	High
R03	Image quality/input variation	Input/data quality	4	3	12	High
R04	Human over-reliance/automation bias	Human oversight	3	4	12	High
R05	Out-of-scope use	Purpose limitation	3	3	9	Moderate
R06	Operational performance degradation	Lifecycle monitoring	3	4	12	High
R07	Insufficient user understanding	Human factors	3	3	9	Moderate
R08	Model overfitting/model selection	Model development	2	3	6	Moderate
R09	Inadequate post-deployment governance	Governance/ accountability	3	4	12	High
R10	Data governance/dataset accountability	Data governance	3	3	9	Moderate
R11	AI model/application integrity	System integrity	2	4	8	Moderate

6. Priority Risk Analysis & Treatment

Priority risk	Inherent	Main treatment	Primary KRI	Residual*
R01 - False-negative classification	16 High	Additional field validation; false-negative analysis; human escalation; user warning; field FNR monitoring.	Field FNR = FN/(FN+TP)	Moderate
R02 - Geographic generalisation	12 High	Define deployment scope; validate new regions; compare location performance; restrict unsupported deployment.	FNR by location	Moderate
R03 - Image/input quality	12 High	Capture guidance; safe Unknown/rejection; retake workflow; device/condition testing.	Unknown/rejection rate	Moderate
R04 - Human over-reliance	12 High	Decision-support designation; limitations; output education; escalation; comprehension testing.	Comprehension failure rate	Moderate
R06 - Performance degradation	12 High	Baseline metrics; new field evaluation; trend monitoring; investigate before retraining; revalidate changes.	Field FNR / recall trend	Moderate
R09 - Post-deployment governance	12 High	System owner; monitoring; incidents; reporting; change	Monitoring completion / open incidents	Moderate

		control; authority to restrict/suspend.		

*Residual ratings are provisional until implementation and control-effectiveness evidence are available.

Human oversight model

Scenario	Human responsibility
Normal prediction	AI informs; human considers the result in context.
Unknown	Retake image or seek assistance; Unknown is not Healthy.
AI/field disagreement	Do not automatically accept AI; retake and/or escalate to agricultural expertise.

Moderate-risk treatment

Risk	Treatment focus	Provisional residual
R05 Out-of-scope use	Purpose limitation + governance before scope expansion	Low-Moderate
R07 User understanding	Onboarding + visual guidance + comprehension testing	Low-Moderate
R08 Model selection	Validation + acceptance criteria + versioning	Low-Moderate
R10 Data governance	Provenance + rights + versioning + lineage + QA	Low-Moderate
R11 System integrity	Release/version control + integrity + access/change control	Low-Moderate

7. AI Monitoring & Oversight Plan

Monitoring should cover the AI system, not model accuracy alone.

ID	Metric / KRI	Domain	Proposed frequency
M01	Field false-negative rate	Model reliability	Monthly
M02	FAW recall	Model reliability	Monthly
M03	Unknown/rejection rate	Input quality	Monthly
M04	AI-field disagreement rate	Human/model interaction	Monthly
M05	Performance by location	Generalisation	Quarterly
M06	User comprehension failure rate	Human oversight	Quarterly
M07	Reported out-of-scope use	Purpose limitation	Quarterly
M08	Open AI incidents	Governance	Monthly
M09	Overdue corrective actions	Governance	Monthly
M10	Unapproved material changes	Change control	Per release + review
M11	Data lineage exceptions	Data governance	Per model release
M12	Integrity verification failures	System integrity	Per release/update

Threshold model

Green: within approved limits - continue routine monitoring. Amber: meaningful deterioration/control weakness - investigate and increase monitoring. Red: approved risk boundary breached or unacceptable risk - escalate and consider restriction. Numeric thresholds should be evidence-based, not invented.

8. Incident, Change & Evidence Governance

AI incident workflow

Detect/report -> Log -> Assess severity -> Verify -> Identify affected model/data/app/location -> Contain -> Root-cause analysis -> Correct -> Validate -> Approve continued/restricted deployment -> Document lessons -> Reassess risks.

Severity	Description
S1 - Minor	Isolated issue with limited consequence
S2 - Moderate	Repeated issue requiring investigation
S3 - Major	Material performance/control failure
S4 - Critical	Systemic failure with potentially serious impact

Material change workflow

Change request -> Impact assessment -> Risk review -> Technical validation -> Governance approval -> Deployment -> Enhanced monitoring.

Material changes include retraining, significant new data, preprocessing/classification changes, Unknown handling changes, geographic expansion, new crops/pests, architecture replacement or material UI changes.

9. Data & Model Governance

Dataset Register - proposed

Field	Example
Dataset ID	DS-001
Dataset	Corn FAW maize-leaf dataset
Purpose	Model development/evaluation
Records	2,868 images
Classes	Healthy / FAW-damaged
Source	Makerere AI Lab / NaCRRI Lacuna project
Geography	Bugiri and Kasese
Collection	ODK
Dataset version	To be established
Data-use rights	Verify/document
Label QA	Document
Data steward	Assign
Associated models	VGG16, InceptionV3, MobileNetV2
Retention	Establish

Model Registry - proposed

Field	Example
Model ID	MDL-001
Architecture	MobileNetV2
Purpose	FAW damage classification
Framework	TensorFlow/Keras
Deployment format	TensorFlow Lite
Associated dataset	DS-001
Model version	To be established
Dataset version	To be established
Validation status	Record
Approval date/by	Record
Deployment version	Record

Integrity/hash	Record
Status	Development / Approved / Retired

Traceability objective
A production artifact should be traceable as: model version -> dataset version -> validation evidence -> approval -> deployed application version.

10. Governance Roles & Accountability

Role	Responsibility
AI System Owner	Overall accountability and deployment decisions
ML/Technical Owner	Model performance and technical changes
Data Owner/Steward	Dataset quality, provenance and lineage
Agricultural Domain Expert	Domain interpretation and escalation
Product/Training Owner	User interaction and competency
Governance/Risk Reviewer	Risk challenge, controls and incidents
Authorized Decision-Maker	Continue, restrict or suspend operation

In a small organisation, one person may perform multiple roles, but responsibilities and decision rights should remain explicit.

11. NIST AI RMF Mapping

Function	Application to Corn FAW Detector
GOVERN	Accountability, oversight, risk register, incidents, change control, data/model registers and evidence.
MAP	Purpose, intended/unsupported use, stakeholders, operating context, human-AI interaction and potential harms.
MEASURE	Model and field evaluation, FNR/recall, Unknown rate, regional performance, disagreement and control-effectiveness monitoring.
MANAGE	Risk prioritisation, treatments, ownership, escalation, residual risk and continue/restrict/suspend decisions.

Lifecycle principle
Assess -> Decide -> Monitor -> Learn -> Reassess.

12. Key Findings & Recommendations

Key findings

- High model accuracy does not eliminate operational risk.
- Failure modes should be prioritized according to their consequences, not aggregate accuracy alone.
- AI governance extends beyond the model to users, inputs, deployment context, data lineage, application integrity and organizational processes.
- Human oversight must define responsibilities, escalation conditions and authority.
- Safe uncertainty can be preferable to forced prediction.
- Monitoring is only useful when connected to thresholds, ownership, escalation and corrective action.
- Governance must continue after deployment.

Priority recommendations

- Conduct additional field validation focused on FAW false negatives.
- Establish explicit approved deployment scope and position the system as decision support.
- Implement input-quality guidance, safe rejection and human escalation.
- Establish Dataset and Model Registers with model-data-release traceability.
- Implement continuous performance/risk monitoring and AI incident management.
- Implement formal model/data/application change control.
- Assign documented authority to continue, restrict or suspend deployment.
- Reassess risks after material changes or significant incidents and retain governance evidence.

13. Governance Evidence Trail

Audit trail

Dataset Register -> Model Registry -> Validation Report -> AI Risk Register -> Risk Treatment Plan -> Deployment Approval -> Monitoring Reports -> Incident & Corrective-Action Register -> Change Records -> Periodic Governance Decision.

Conclusion

The Corn FAW Detector demonstrates how technical machine-learning evidence can be translated into structured AI governance. The original MSc project established a functioning computer-vision application and generated technical and field evidence. This retrospective assessment extends that work across model reliability, data, human interaction, purpose limitation, monitoring, governance and system integrity.

Responsible deployment cannot be established through predictive accuracy alone. Effective governance requires understanding system purpose, operating context, affected stakeholders, failure pathways, human interaction, data provenance, technical controls and organizational accountability.

NIST AI RMF summary

GOVERN the system and accountability. MAP the context, stakeholders and harms. MEASURE technical and operational risk. MANAGE risks through controls, monitoring and evidence-based decisions.

Appendix A - AI Incident Record Template

Field	Entry
Incident ID	
Date detected	
Reporter	
System/model version	
Application version	
Dataset version (if relevant)	
Location/context	
Description	
Severity	
Verification status	
Affected users/scope	
Immediate containment	
Root cause	
Corrective action	

Action owner	
Risk register affected?	
Revalidation required?	
Closure date	
Final governance decision	

Appendix B - AI Change Record Template

Field	Entry
Change ID	
Proposed change	
Reason	
Model/data/application affected	
Material change?	
Risks affected	
Impact assessment completed	
Validation required	
Validation result	
Approved by	
Deployment date	
Enhanced monitoring required	
Post-change outcome	

Appendix C - Periodic AI Governance Report

- System and deployment status; current model/application versions.
- Performance trends, including false negatives, recall and Unknown/rejection.
- Geographic performance and user-understanding findings.
- Incidents, corrective actions and material system/data changes.
- Control-effectiveness findings and changes to risk ratings.
- Explicit deployment recommendation: Continue; Continue with additional controls; Restrict; or Temporarily suspend pending remediation.

Portfolio Disclaimer

This case study is a retrospective AI governance assessment based on a system originally developed as academic research. Technical results identified as originating from the MSc project reflect the original research evidence. Governance structures, risk scores, treatment plans, monitoring mechanisms and control recommendations identified as retrospective or proposed were developed subsequently for this AI Governance portfolio and should not be interpreted as controls necessarily implemented during the original MSc study.